COS80001 - Cloud Computing Architecture

Assignment 1 - Part B

Creating and deploying Photo Album website onto a simple

AWS infrastructure

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Submitted to: Dr. Ammara Khan

Creating and deploying Photo Album website onto a simple

AWS infrastructure

Task 1: Create a secure Virtual Private Cloud (VPC) with subnets, routing tables and security groups.

Task 2: Creating Instance and Instance Server

Task 3: Creating Database

Task4: Access to and from VPC via internet gateway

Task 5: Network ACL

Task 6: Uploading Images in S3 bucket

Task 7: Testing Webserver in PHP website

Task 8: Display Photo

COS80001

Cloud Computing Architecture

Assignment 2

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Developing a highly available Photo Album website

URL:

1.0 Functional Requirement of Photo Album Website:

1.1Photo album (album.php)

Making Sure Assignment 1b is working properly.

Afterward, new photoalbum.php file was downloaded and uploaded to var/www/html/folder. According to assignment 1b values in constants.php are changed.

Checking if the website works well.

1.2 Photo Uploading:

This page allows to upload a photo to an S3 bucket and inserting metadata into RDS.

2.0 Infrastructure deployment:

I verified existing VPC and using the link provided I created NAT instance. I have created a security group NATSG where own VPC is selected and three inbound rules are added which are HTTP, HTTPs and SSH.

I created a Instance name NATinstances.

I selected NATSG security group, choosing the given AMI.

I selected my own VPC, enabled auto-assign IP, enabled public Subnet 10.0.1.0/24 and selected existing security group NATSG.

Then after I launched Instance:

2.2 S3 Photo Storage:

This part was already created in Assignment 1b.

2.3 Load Balancing:

I created Image with name AssignmentAMI.

I created load balancer named AssignmentELB

Creating Target group using correct Path.

2.3 NAT:

I have creeated a NAT instance in the previous step in 2.1 by verifying VPC.

2.4 Auto-Scaling:

With the AMI I created I lunched configuration and selected IAM instance profile as LabinstanceProfile, and WebServerSG.

Two private subnets are selected.

Grouping size and scaling policies are configured.

Finally, I created Auto Scaling Group.

2.5 EC2 instances:

I created a Dev Server instance using DevSecurityGroup.

2.6 Creating Thumbnail Lambda Function:

2.7 Database with RDS:

I didnt change anything as no changes were required for this task.

2.8 Secuirty Groups:

ECLSecurityGroup is added for AssignmentELB.

2.9 Network ACL:

I created PrivateSubnetNACL and PublicSubnetNACL where I edited inboud rules adding ICMP rules that block DevServer requests in PrivateSubnetNACL while I edited outbound rules in PublicSubnetNACL by implementing ICMP rule to deny DevServer in incoming traffic.

Testing:

Checking if the resized photographs exist in S3 or not.

I could successfully upload photos with modified details from constants.php

This is the link to access the album.php:

While trying through NAT Instance, it didnt allow to access. Then new auto-created instances were immediately re-created after being terminated.